

Content, Not Format: What Drives Analysis-Mode in Language Model Agents, and How Grounding Recovers Execution

Sophie D. Neilson
Independent Researcher
ORCID: [0009-0001-2430-1743](https://orcid.org/0009-0001-2430-1743)

September 2026

Abstract

A prior report observed that a small language model, when a task is prefixed with a structured description of a recurring decision, stops acting and instead analyzes the description. That study used one model and one scenario, and it did not separate the description’s content from its form. This follow-up adds the controls it lacked. We prefix a task with either a three-axis descriptive “glyph” or an information-matched imperative rule, across four decision classes, and we sweep the descriptive prefix across four open-weight models on the original class. The shift away from action generalizes across models, and it is a content effect, not a format effect: the matched imperative suppresses action as much as the glyph, and the descriptive format shows no consistent advantage over a plain rule with the same content. Control prefixes (a content-scrambled glyph, a wrong-class glyph) show the suppression mechanism is decision-class-dependent. We then vary two things independently while holding the content fixed: form (descriptive versus directive) and grounding (a domain-free statement versus a domain-specific one). Grounding is the lever. A domain-specific aid restores execution where a domain-free one suppresses it, in both forms. Whether a domain-specific *description* suffices as well as a domain-specific *command* is under-determined here: the command reaches ceiling in all four model-and-class cells, so the two separate only where the description falls short of ceiling, which happens once, on the smaller model and the harder task, and there the command does better. All runs use one local raw-completion inference rig; reproducing them needs a 24 GB GPU and no paid interface. We state the scope and the measurement limits, including the cells where a baseline already acts and no effect is measurable.

1 Introduction

A prior report [Neilson, 2026] documented a dissociation between comprehension and execution in Mistral-7B-Instruct. When the model received a task scenario prefixed with a *glyph*, a three-axis structured description of a recurring decision class, it analyzed the description rather than acting on the task. Across three legs of eight runs it produced no task execution under the glyph, while producing the target action in most runs with no glyph present. The report named this the *analysis-mode register shift* and scoped the observation to one glyph, one scenario, and one model. It named format cueing as one of three candidate explanations and disclaimed a mechanism.

That report named two open questions in its own limitations. The first is breadth: whether the shift is specific to Mistral-7B. The second is the harder one. The glyph carries information about the decision, and it also carries a particular form: named axes, written descriptively rather than as instructions. The study compared a glyph against no glyph, never against a

different delivery of the same information, so it could not say whether the descriptive *format* drove the shift or whether the *content* did, by whatever route it arrived.

This follow-up runs the missing controls and then goes past them. We hold the information constant and vary the delivery. For each decision class we write an *imperative*: a set of directives stating the same failure, correct action, and applicability boundary as the glyph, in commanding rather than descriptive form, audited for content parity. We run the glyph across four open-weight models. We add two mechanism controls that damage the glyph in one way each: a *scrambled* glyph keeps the three-axis shape but destroys the content, and an *off-target* glyph is coherent but describes a different decision class. Finally we cross two factors while holding content fixed: form (descriptive versus directive) and grounding (a domain-free statement versus a domain-specific one).

The picture that results refines the prior observation rather than overturns it. The register shift is real and general. It is not a property of the descriptive format. The content the glyph carries, present equally in a matched rule, is what moves behavior; which part of the content matters depends on the decision class; and what recovers execution is grounding the decision in the concrete domain.

1.1 Terminology

These terms come from the research program this study belongs to, not from the field’s shared vocabulary. Each is defined here at first use.

- **Glyph**: a three-axis structured description of a recurring decision class, written from inside the decision. Its axes are *Marker* (what the failure looks like), *Aim* (what correct navigation looks like), and *Rest* (when the class does not apply). A glyph describes; it does not instruct. It is domain-free: it names abstract roles, not a specific tool or file.
- **Decision class**: a recurring situation in which an agent can act correctly or fail in a characteristic way, described apart from any one scenario.
- **Analysis-mode register shift**: the pattern in which a model, given a glyph, discusses the decision class rather than acting within it. It recognizes the obligation and does not carry it out.
- **Imperative**: a directive rewriting of a glyph’s content. It states the same propositions as commands rather than as description, and stays domain-free. It is the content-and-form control.
- **Ground**: a domain-specific statement of the decision for the concrete scenario, in descriptive form (it says what a correct outcome is, without naming a command and without pasting the finished artifact).
- **Domain-directive**: the same domain-specific content as the ground, in directive form (it commands the action). Ground and domain-directive differ only in mood.

1.2 Related Work

Two recent papers occupy adjacent territory. Wang et al. [Wang et al., 2026] study narrative priors: they run structurally identical games under different storylines and find that the task narrative explains more behavioral variance than an assigned persona does. Their work shows framing context shifts behavior. It does not test a descriptive description of a decision class, and it does not separate comprehension from execution. Liu et al. [Liu et al., 2026] derive behavioral mode axes, activation-space directions from contrastive behavioral traces, and use them to measure and steer a model’s behavioral style. Their notion of a behavioral register is

relevant to the analysis-mode phenomenon, though they steer through activations rather than a prompt-level description.

The finding also sits next to a broader literature on prompt-format sensitivity and on instruction-following in language models, which studies how the surface form of a prompt changes behavior. We did not run a systematic survey of that literature, and we make no universal novelty claim; a full positioning against it is future work. The distinction this study turns on, description versus instruction, maps loosely onto Ryle’s *knowing-that* versus *knowing-how* [Ryle, 1949]. The theory behind the glyph draws on Polanyi’s account of tacit knowledge [Polanyi, 1966] and Klein’s recognition-primed decision model [Klein, 1998], under which an agent that recognizes a decision class from prior exposure navigates it without explicit rules. The present result bears on that theory, and we return to it in the discussion.

2 Materials and Method

2.1 Decision Classes

We test four decision classes, each a domain-free description instantiated in one task scenario.

- **casg-direct**: an agent finishes a software release but leaves the companion `CHANGELOG.md` not updated. Correct action: add the release entry. This is the class from the prior report.
- **formal-step**: an agent must produce a compliance checklist by working it from the current specification, not by carrying a prior sign-off across. Working it surfaces a control the prior summary omits.
- **parent-state**: an agent must check a parent or milestone state before acting on a task.
- **conditional-gate**: a conditional gate that most models already handle unprompted in the tested scenario.

2.2 Conditions

The core study crosses two factors while holding the propositional content fixed, giving a two-by-two plus a baseline:

- **baseline**: the scenario alone.
- **glyph**: descriptive, domain-free (the three-axis glyph).
- **imperative**: directive, domain-free (a rule with the same content).
- **ground**: descriptive, domain-specific.
- **domain-directive**: directive, domain-specific.

The four aids carry the same core proposition at two grounding levels; a content-parity audit records that the glyph and the imperative state the same facts, that the ground and the domain-directive state the same facts and differ only in mood, and that neither the ground nor the domain-directive hands over the finished artifact. Two mechanism controls take the glyph’s slot: a *scrambled* glyph (three-axis shape kept, words shuffled within each axis) and an *off-target* glyph (a coherent glyph for a different class).

2.3 The No-Tools Notice

The subject models are agentic: given a task, a strong model often emits tool calls rather than inline prose. The probe runs in a single-turn harness that returns nothing to a tool call, so a tool-calling reply stalls and produces no scoreable action. That stall is a harness artifact rather than an analysis-mode response. We append a short notice to the user turn, stating that no tools are available and to answer inline. The notice is identical across all conditions, so it cannot bias the contrasts, though it plausibly nudges every condition toward inline prose. Under the notice the grids ran with no tool stalls.

2.4 Models and Sampling

The two-by-two ran on Qwen3.8-27B-Instruct (Q4_K_M) and Mistral-7B-Instruct-v0.3 (Q4_K_M), n=24 per cell. The four-class content grid and the mechanism controls ran on the same two models at n=8 per cell. The cross-model sweep added phi-4-14B and Qwen2.5-32B-Instruct on casg-direct at n=8.

All subjects are served by one local server (`llama-server`, `llama.cpp`) over a raw completion path with a minimal per-model instruct wrapper, thinking disabled, and no system prompt. One model runs at a time, verified from the server’s properties endpoint on each swap. Running every subject on this one local rig is a reproducibility decision: the results reproduce on a single 24 GB GPU with no paid interface. A hosted model cannot run on this path, so no hosted model is a subject here (Section 2.5).

The complete sampler chain, identical across conditions and models: temperature 0.8, top_k 0, top_p 1.0, min_p 0.05, all penalties disabled, maximum 1024 new tokens (raised to 1536 for formal-step in the two-by-two, whose replies are longer; the four-class grid and the mechanism controls used 1024), seeds 1 through the run count. Penalties are disabled by design: a repetition penalty suppresses repeated structure, and axis-by-axis analysis is repeated structure, so a live penalty would attenuate the behavior under study. Qwen2.5-32B was served at a reduced 8K context to fit the GPU; prompts are about 5K tokens. Each run records the served model, the build id, the throughput, the sampler read back from the server, the image content digest, the substrate, and the repository commit. The two-by-two ran on image `sha256:67d26e73`; the mechanism controls on `sha256:e159c2b8`; the four-class grid on `sha256:82182b74`; the sweep’s new arms (phi-4, Qwen2.5-32B) on `sha256:e159c2b8` and its Mistral and Qwen3.8 arms on `sha256:82182b74`. Because the sweep pools cells from two images, one of which followed a fix to a reply-truncation bug, we read the sweep as cells and direction, not pooled counts; its replies are short, so the truncation fix does not bear on that class.

2.5 Scoring

Responses are scored with a five-code rubric: **C**, recognition and the correct action; **Ii**, recognition without action; **Ic**, recognition with a defective action; **I**, no recognition; **N**, not scoreable. The correct-action bar is identical across the glyph, imperative, ground, and domain-directive conditions, so no scoring asymmetry can confound the contrasts. We report an execution rate (an artifact produced: C or Ic) alongside the strict correct-target rate (C), because the load-bearing distinction is whether the model acts at all versus analyzes without acting. The per-class rubrics are in Appendix A, and the strict correct-target table is Table 4.

Each cell was scored by two independent raters, both Claude (Anthropic) subagents, blind to the condition and to the predictions, then adjudicated by the analyst. Claude is a permitted judge but never a subject: it cannot run on the local raw-completion rig that the reproducibility apparatus requires. The two-by-two (480 responses) reached a pooled pre-adjudication agreement of 477 of 480 (99.4%); the three disagreements, all in one model-and-class cell, were adjudicated against the rubric. The predictions were written before the runs and kept out of

everything the subject and the rater saw. Two raters drawn from one model measure consistency, not validity, because they share a training distribution. To test validity, an author (a person) independently blind-scored a stratified 40-response subset, weighted to the load-bearing casg-Mistral ground cell, blind to the condition and to the automated codes. Human-to-Claude exact-code agreement was 39 of 40 (97.5%), and 23 of 24 (96%) on the load-bearing cell, where the human counted 12 executions to the automated 13. The Claude scoring is therefore validated against a human anchor, and the conclusions rest on gaps far larger than the one-run human-machine difference or any plausible rater slip. Table 1 lists the raters.

2.6 Agents and Models

Table 1: Agents and models. “Not recorded” means the study record does not hold the version.

Role	Model	Version and runtime	Date
Subject (two-by-two, grid, controls)	Qwen3.8-27B-Instruct;	Q4_K_M GGUF, llama.cpp raw completion, digest-pinned	2026-09-08;
	Mistral-7B-Instruct-v0.3	image	2026-09-14
Subject (sweep)	phi-4-14B; Qwen2.5-32B-Instruct	Q4_K_M GGUF, llama.cpp; Qwen2.5-32B at 8K context	2026-09-09
Raters, two-by-two	Claude (Anthropic) subagents	Not recorded. Two blind raters per cell; adjudicated by the analyst.	2026-09-14
Raters, grid / sweep / controls	Not recorded	Blind subagents, model not named in those records.	2026-09-08;
Human rater (validity anchor)	Author (a person)	Blind, stratified 40-response subset; 97.5% exact-code agreement with the Claude raters.	2026-09-14

3 Results

3.1 The Register Shift Generalizes Across Models

Table 2 reports the correct-action count for casg-direct across four models. Every model that acts unprompted at baseline moves away from action under a glyph: Mistral 8 to 0, phi-4 8 to 0, Qwen3.8 8 to 4. On Mistral the suppressed runs split evenly between recognition-without-action and no-recognition; on phi-4 they are all recognition-without-action. The direction reproduces in three model families and four sizes, so the analysis-mode shift is not specific to Mistral-7B.

Table 2: Correct-action count (out of 8) by model and condition, casg-direct. Non-C codes for the glyph and imperative are in parentheses.

Model	Baseline	Glyph	Imperative
Mistral-7B	8	0 (4 Ii, 4 I)	1 (7 Ii)
phi-4-14B	8	0 (8 Ii)	0 (8 Ii)
Qwen2.5-32B	4 (4 Ii)	2 (4 Ii, 2 Ic)	6 (2 Ic)
Qwen3.8-27B	8	4 (4 Ii)	4 (4 Ii)

The same table gives the content answer for this class. The information-matched imperative suppresses action as much as the glyph or more: identical at 0 for phi-4, identical at 4 for

Qwen3.8, near zero for Mistral, and, for Qwen2.5-32B, the plain imperative reaches 6 where the glyph reaches 2. On this class the descriptive format never beats the matched rule.

3.2 Content, Not Format, in the Measuring Cells

The four-class content grid (n=8) extends the contrast to parent-state and conditional-gate. The reading needs a calibration caveat: on the strong model the baseline is at ceiling for casg-direct, formal-step, and conditional-gate, so those cells cannot measure a guidance effect. Five of the eight class-and-model cells are at ceiling; the contrast is measurable in three: parent-state on both models and conditional-gate on Mistral. In them the glyph never beats the imperative. On parent-state (Qwen) both lift correct action from a baseline of 1 of 8 to 8 of 8, identically. On conditional-gate (Mistral) the imperative reaches 5 of 8 and the glyph 2. On parent-state (Mistral) the imperative reaches 7 of 8 and the glyph 1, because the model used the glyph’s non-firing clause to reason itself out of the check. The register shift measured as recognition without action is also not format-specific: it appears under the glyph and the matched imperative at similar rates (for example, casg on Qwen, 4 versus 4; formal-step on Qwen, 4 versus 5). The four-model sweep of Section 3.1 is casg-direct only, so the breadth is four models on one class and four classes across three measuring cells, not a full grid.

3.3 The Suppression Mechanism Is Class-Dependent

Two control glyphs test what about the glyph does the work. A scrambled glyph keeps the three-axis shape and destroys the content; an off-target glyph keeps coherence and changes the class. On casg-direct both controls reproduce the shift (on Qwen the scrambled glyph gives 5 recognition-without-action and the off-target 4, against the real glyph’s 4 and a baseline of 1), so on that class neither comprehension nor scenario recognition is necessary; the bare structured prefix carries it. That reading does not generalize. On formal-step the scrambled glyph does not reproduce the shift (the model acts, like baseline) while the off-target one does, so comprehension is load-bearing there. On parent-state the off-target glyph fails to cue the check while the scrambled one still cues it, so recognition is load-bearing there. Across the informative classes, either comprehension or recognition carries the effect, and not the same one. The honest statement is that the glyph’s suppression is class-dependent, and the control that fails names the ingredient the class relies on. This walks back the single-class reading (that mere structure suffices), which held only on casg-direct.

3.4 Grounding Recovers Execution

The two-by-two (n=24) varies form and grounding with the content fixed. Table 3 gives the execution rate. At n=24 a 95% Wilson interval on a proportion is roughly ± 0.18 wide, so a gap smaller than about 5 of 24 is within noise, and a gap of 10 or more of 24 clears it. In every cell the domain-free aids (glyph, imperative) suppress execution below baseline, and the domain command (domain-directive) restores it to ceiling. The descriptive ground restores execution relative to the domain-free aids in every cell (for example casg-Mistral, glyph 0 and imperative 5 rise to ground 13), and it reaches ceiling on three of four cells. The property that flips a prefixed decision-aid from suppressing action to permitting it is grounding: a domain-free aid suppresses, a domain-specific aid recovers, in both descriptive and directive form.

Whether a domain-specific *description* suffices as well as a domain-specific *command* is under-determined by this design. The domain-directive reaches ceiling (24 of 24) in all four cells, so the two aids can be separated only where the ground falls short of ceiling. That happens in one cell, casg on Mistral, and there the command (24) beats the description (13) by a gap that clears the noise band. In the other three cells both aids are at joint ceiling and cannot be distinguished. One piece of evidence points the other way: on casg-Qwen the descriptive ground

Table 3: Execution rate (an artifact produced: C or Ic, out of 24) in the form-by-grounding design.

	casg Mistral	casg Qwen	formal Mistral	formal Qwen
baseline	19	23	24	22
glyph (descriptive, domain-free)	0	15	14	16
imperative (directive, domain-free)	5	17	10	17
ground (descriptive, domain-specific)	13	24	24	24
domain-directive (directive, domain-specific)	24	24	24	24

reaches the strict correct-target bar in 24 of 24 runs against the command’s 17 (Table 4), so the description produced better-formatted output there. Taken together, a grounded description clearly recovers execution, but the claim that it does so *as well as* a command is suggested, not established: the sole discriminating cell favors the command.

4 Discussion

4.1 The lever is grounding, not format or form

The prior report named the glyph’s descriptive format as one candidate cause of the register shift. Under control that candidate does not carry the effect. A domain-free imperative, the same content in directive form, induces the shift as strongly as the glyph, so form is not the lever. Damaging the glyph shows the suppression runs through different ingredients in different classes. And restoring execution turns on one thing across every cell: whether the aid is grounded in the concrete domain. An abstract decision-description, however phrased, pulls the model toward analyzing the decision; grounding the same decision in the domain keeps the model acting on it. The cleanest single line of evidence is the prose-versus-prose contrast, imperative to domain-directive, which holds form constant and adds only grounding: execution rises 5 to 24, 17 to 24, 10 to 24, and 17 to 24 across the four cells.

4.2 What this says about comprehension and compliance

The recognition-primed account predicts that a model recognizing a decision class navigates it without explicit rules. This study can say that a grounded description, which commands nothing, recovers execution well above the domain-free aids, and reaches ceiling in three of four cells. It cannot say that a description suffices as well as a command, because the command is at ceiling everywhere and the two separate in only one cell, where the command wins. So grounded comprehension is enough to recover much of the execution the abstract aids suppressed, and whether it fully matches a directive is open, with the one decisive comparison favoring the directive and one strict-format comparison favoring the description. We do not resolve it here.

4.3 Relation to the prior report

This study extends the prior report rather than corrects it. The register shift the report described is real and now shown across four models. What the report could not see, for want of a content control, is that the effect was not carried by the descriptive format. An earlier probe in the same program also reported that adding a ground recovers execution, but that probe added the ground *alongside* the glyph and its ground pasted the finished artifact, so the recovery could have been the model copying an answer. Here the ground stands alone and names the action without writing it, and the recovery survives, so the ground effect is neither a copy artifact nor dependent on the glyph. What the de-confounding changes is the mechanism: grounding is the lever.

4.4 Limitations

1. **Scope.** The two-by-two covers two classes and two models; the breadth results add two models on one class and two more classes at $n=8$, measurable in three cells. Each class has one scenario and one authored glyph. A different scenario or glyph could behave differently.
2. **Small samples and ceilings.** Cells are 8 or 24 runs, and several baselines and the domain command are at or near ceiling, which is why the description-versus-command comparison separates in only one cell. We read direction and gaps that clear the noise band, not single integers.
3. **Rater validity anchored on one human, not a second model family.** Both automated raters were Claude subagents, blind and adjudicated, with 99.4% pooled agreement (consistency). Their validity was checked against a human anchor: an author blind-scored a stratified 40-response subset and agreed with the automated codes on 39 of 40 (97.5%), and on 23 of 24 in the load-bearing cell. The human anchor is a single author rather than several independent people or a second model family, and it covers a subset, not the full grid; the conclusions rest on gaps far larger than the residual disagreement (for example 0 or 5 rising to 24).
4. **A residual in one contrast.** The glyph is a three-axis block and the other aids are prose, so the glyph-versus-ground contrast mixes grounding with block structure. The imperative-versus-domain-directive contrast (prose versus prose) is the cleaner grounding read and shows the same thing; the mechanism controls bound the structure question separately.
5. **No mechanism below the behavioral level.** We report which ingredient is necessary per class and which delivery recovers execution. We do not claim to know why a prefixed description cues analysis inside the model.

5 Conclusion

A structured description of a decision class prefixed to a task suppresses execution into analysis across four open-weight models. Under an information-matched control the effect is content, not the descriptive format; a plain rule with the same content suppresses as hard. Control prefixes show the suppression is class-dependent, resting on bare structure, on comprehension, or on scenario recognition in different classes. Execution is recovered by grounding the decision in the concrete domain, in both descriptive and directive form. Whether a grounded description recovers execution as fully as a command is left open by ceiling effects, with the one decisive cell favoring the command. We report the finding at the scope of the tested glyphs, scenarios, and models, and invite replication on the one local rig the study runs on.

5.1 Data Availability

The complete study materials (scenarios, glyphs, imperatives, grounds, scoring rubrics, per-run scores, run records, and control materials) are available at <https://github.com/sophdn/corpos-lab> under the `studies/matched-content-experiment` directory.

References

Klein, G. A. (1998). *Sources of Power: How People Make Decisions*. MIT Press.

- Liu, H., Liu, R., Xu, H., Han, J., Fang, S., Yan, S., Deng, H., Wang, G., and Zou, N. (2026). Your LLM, Your Style: Behavioral Mode Axes for LLM Behavioral Control. *arXiv preprint arXiv:2608.10703*.
- Neilson, S. D. (2026). Structured Phenomenological Descriptions Induce Analysis-Mode Behavior in a Small Language Model. *Zenodo*. [doi:10.5281/zenodo.22542747](https://doi.org/10.5281/zenodo.22542747).
- Polanyi, M. (1966). *The Tacit Dimension*. Doubleday.
- Ryle, G. (1949). *The Concept of Mind*. Hutchinson.
- Wang, Y., Lester, J., and Srivastava, S. (2026). The Story Shapes the Agent: Narrative Priors in LLM Behavior. *arXiv preprint arXiv:2607.18566*.

A Scoring Rubrics

The correct-action bar is condition-independent within a class. For casg-direct, C is a `CHANGELOG.md` entry for v1.5.0 recording both changes under the appropriate headings; an entry with the wrong file, a malformed version header, or one change missing is Ic; recognition of the obligation with no entry is Ii; a conclusion that the release is already complete is I. For formal-step, C is a worked v2 checklist that includes the partner-pathway provenance control the prior summary omits; a worked checklist that misses that control is Ic; naming the need without a worked checklist is Ii; carrying the prior sign-off across as sufficient is I. The full rubrics are committed with the study.

B Strict Correct-Target Rate

Table 4 gives the strict correct-target rate (C only) for the two-by-two, alongside the execution rate of Table 3. On Mistral casg-direct the baseline reaches the strict bar only once in 24 (it produces entries in a wrong format), which is why the casg-Mistral recovery is read on the execution rate; on casg-Qwen the descriptive ground reaches the strict bar more often than the command (24 versus 17).

Table 4: Strict correct-target rate (C only, out of 24).

	casg Mistral	casg Qwen	formal Mistral	formal Qwen
baseline	1	22	24	22
glyph	0	15	14	15
imperative	0	16	10	17
ground	12	24	24	24
domain-directive	24	17	24	24

C Materials

The scenario, glyph, imperative, ground, and domain-directive for each class are reproduced in the repository under `studies/matched-content-experiment`. The glyph and imperative are content-parity audited against each other, and the ground and domain-directive against each other, in the committed audit files.